

Tao2.7

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1 Outer Measure

For any $A \subset \mathbb{R}^n$ define $m^*(A) = \inf \{ \sum_{i \in I} \text{vol}(B_i) : \{B_i\}_{i \in I} \text{ is a countable cover of } A \}$. If the sum diverges, take the sum to be ∞ .

I claim that m^* is well-defined. It suffices to show there exists a countable cover of $\mathbb{R}^n$.

Proof: Let $B_k = (-k, k)^n$ for all $k \in \mathbb{Z}^+$. Suppose $x \in \mathbb{R}^n$. There exists $i \in [n]$ such that $|x_i|$ is maximal. By the Archimedean principal, pick $k > |x_i|$, so $x \in B_k$ since $|x_i| < k$ for all $i \in [n]$. It follows $\bigcup_{k \in \mathbb{Z}^+} B_k = \mathbb{R}^n$.

If $A \subset B$, then $m^*(A) \leq m^*(B)$, since a cover of B is also a cover of A .

I claim that $m^*(\bigcup_{i \in I} A_i) \leq \sum_{i \in I} m^*(A_i)$ for any countable collection $\{A_i\}$ of subsets of $\mathbb{R}^n$.

Proof: Let $\mathcal{B}_i$ denote a cover of A_i . Then $\mathcal{B} = \bigcup_{i \in I} \mathcal{B}_i$ is a cover of $\bigcup_{i \in I} A_i$, so

$$\sum_{i \in I} m^*(A_i) = \sum_{i \in I} \sum_{B \in \mathcal{B}_i} \text{vol}(B) \geq \sum_{B \in \mathcal{B}} \text{vol}(B) \geq m^*\left(\bigcup_{i \in I} A_i\right).$$

Take the empty cover to have zero sum, so $m^*(\emptyset) = 0$.

I claim that $m^*(I^n) = 1$. Let $B_k = (-\frac{1}{k}, 1 + \frac{1}{k})^n$. Note that $\{B_k\}$ is a cover of I^n and $\text{vol} B_k = (1 + \frac{2}{n})^n$, and $\lim_{k \rightarrow \infty} \text{vol} B_k = (\lim_{k \rightarrow \infty} 1 + \frac{2}{n})^n = 1$, so $m^*(I^n) \leq 1$.

Let $\mathcal{B}$ be a cover of I^n . I claim that $\sum_{B \in \mathcal{B}} \text{vol}(B) \geq 1$. Note that I^n is compact. By Lebesgue number lemma, pick $\delta > 0$ such that if $\text{diam } C \leq \delta \rightarrow C \subset A$ for some $B \in \mathcal{B}$. By Archimedean principle, pick $m \in \mathbb{Z}^+$ such that $1/m < \delta$. Pick $\epsilon > 0$ such that $1/m + 2\epsilon < \delta$. For each $\alpha \in [m]^n$, let $B_\alpha = \prod_{i \in [n]} \left(\frac{\alpha(i)-1}{m} - \epsilon, \frac{\alpha(i)}{m} + \epsilon \right)$.

Note that $\{B_\alpha\}_{\alpha \in [m]^n}$ is a refinement of $\mathcal{B}$, and covers I^n . It follows $\bigcup_{\alpha \in [m]^n} B_\alpha \subset \bigcup_{B \in \mathcal{B}} B$. We have

$$\sum_{\alpha \in [m]^n} \text{vol}(B_\alpha) = m^n \cdot \left(\frac{1}{m} + 2\epsilon \right)^n = (1 + 2m\epsilon)^n.$$

Taking $\epsilon \rightarrow 0$, we have $m^*(I^n) \geq 1$, so $m^*(I^n) = 1$, as desired.

Translations preserve volume, so $m^*(A) = m^*(x + A)$ for all $x \in \mathbb{R}^n$.

I claim that $m^*(\lambda A) = |\lambda|^n m^*(A)$. If $\lambda = 0$, the result holds. Else, suppose $\mathcal{B}$ is a cover of A . Then $\lambda \mathcal{B}$ is a cover of λA , so $|\lambda|^n m^*(A) \geq m^*(\lambda A)$. Similarly, $|1/\lambda|^n m^*(\lambda A) \geq m^*(A)$, so $|\lambda|^n m^*(A) = m^*(\lambda A)$, as desired.

It follows $m^*(\mathbb{R}^n) \geq m^*([-m, m]^n) = (2m)^n \geq m$ for all $m \in \mathbb{Z}^+$, so $m^*(\mathbb{R}^n) = \infty$.

I claim that $m^*(A) = m^*(\overline{A})$. If $A = \emptyset$, we're done. Assume otherwise. It is enough to show $m^*(A) \geq m^*(\overline{A})$. I claim that $\overline{A} = \bigcap_{m \in \mathbb{Z}^+} U(A, 1/m)$. Suppose $x \notin \overline{A}$, so $d(x, A) > 0$. By Archimedean principle, there exists $m \in \mathbb{Z}^+$ such that $1/m < d(x, A)$, so $x \notin U(A, 1/m)$. Now suppose $x \in \overline{A}$. Then $d(x, A) = 0$, so $x \in U(A, 1/m)$ for all $m \in \mathbb{Z}^+$.